SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

F Y B Tech (Semester I and II)
(Common to all branches)

Teaching, Credit and Evaluation Scheme

From
Academic Year 2025-26
(Version 3.0)

Approvals

- **Scheme of semester I & II and detailed syllabus of semester I** is approved in the common special meeting of the Boards of Studies in Engineering of All Departments jointly with the Faculty of Engineering & Technology conducted on 7 July 2025

- **Scheme of semester I & II and detailed syllabus of semester I** is approved in the Academic Council meeting conducted on 17 October 2025

- **Amendment in the scheme of semester I & II and detailed syllabus of semester II** is approved in the meeting of the Board of Studies in
    - Mechanical Engineering conducted on 12 November 2025
    - Electronics Engineering conducted on 21 November 2025
    - Information Technology and Artificial Intelligence & Data Science conducted on 21 November 2025
    - Electronics and Telecommunication Engineering conducted on 24 November 2025
    - Computer Engineering conducted on 24 November 2025
    - Robotics and Artificial Intelligence conducted on 25 November 2025

- **Amendment in the scheme of semester I & II and detailed syllabus of semester II** is approved in the meeting of Faculty of Engineering and Technology conducted on 27 November 2025

Preamble

(From the Dean, Faculty of Engineering and Technology)

It gives me immense pleasure to introduce the revised curriculum for the First Year B. Tech. (R-2025) at K J Somaiya School of Engineering, designed with a forward-looking vision to shape engineers of the future. This revision is aligned with the guiding principles of the National Education Policy (NEP) 2020, and in compliance with UGC and AICTE guidelines, while incorporating valuable inputs from our Academic Council, faculty members, and industry experts.

The R-2025 curriculum represents a significant step toward holistic and application-oriented learning. It has been thoughtfully structured to provide a strong foundation in engineering sciences, while fostering creativity, critical thinking, and multidisciplinary exposure from the very first year. The introduction of branch-specific core and science courses is a strategic move to immerse students early into their chosen fields. These courses enhance academic interest, improve retention, and allow students to apply scientific principles directly to domain-specific problems from the outset.

A key highlight of the revised curriculum is the Makerspace course, which brings innovation and experiential learning to the forefront. It blends traditional workshop practices with real-world project-building, enabling students to explore design thinking, rapid prototyping, and interdisciplinary collaboration. This hands-on approach is aimed at nurturing problem-solving abilities and an innovation mindset, essential for success in today's dynamic engineering landscape.

A particular emphasis has been laid on developing programming and problem-solving skills through structured and object-oriented programming methodologies, with flexibility in language platforms to suit diverse academic backgrounds and branch-specific needs. The inclusion of end-semester practical examinations and distinct passing criteria for continuous and end-semester assessments further strengthens academic rigor and continuous learning.

The inclusion of value-added courses like Sports, Environmental Science, and Biology for Engineers reflects our commitment to shaping well-rounded individuals equipped not only with technical skills but also with physical well-being, ecological awareness, and interdisciplinary insight. The foundational course in Biology for Engineers will be further complemented with advanced courses in higher semesters, enabling students to understand complex problems in biological sciences and healthcare. These courses will guide them to identify opportunities where engineering and technology can offer impactful solutions.

This curriculum is not just a syllabus; it is a blueprint for nurturing engineers who are resilient, innovative, and ethically grounded. We are confident that the R-2025 curriculum will empower our students to become competent professionals and responsible global citizens, ready to contribute meaningfully to society and the engineering world.

Dr. Suresh Ukarande
Dean, Faculty of Engineering & Technology

About SVU R-2025 Curriculum

1. **Introduction:**
K J Somaiya School of Engineering has revised the First Year B. Tech curriculum in alignment with the National Education Policy (NEP) 2020, UGC and AICTE guidelines, and suggestions from the Academic Council. The R-2025 curriculum aims to foster holistic, multidisciplinary, and application-oriented learning for engineering students.

2. **NEP 2020 Alignment:**
The curriculum reflects NEP 2020 principles by:
   - Promoting **multidisciplinary and holistic education**
   - Emphasizing **skill-based and project-oriented learning** through Makerspace
   - Ensuring **flexibility in learning paths** (language options in programming)
   - Encouraging **physical education** and **environmental awareness**
   - Introducing **value-added, branch-specific, and contextual learning** early in the program

3. **Salient Features of FY B. Tech Curriculum (SVU R-2025):**
   a. **Defined Value-Added Course Basket (Sports):** Introduced as a structured and credit-based course as per NEP-2020 guidelines to promote physical well-being and teamwork.
   b. **Environmental Science (EVS):** Added as a compulsory course as per UGC and AICTE mandates, addressing sustainability, ecological literacy, and climate challenges.
   c. **Biology for Engineers:** Included based on the Academic Council's recommendation to promote interdisciplinary thinking and prepare students for bio-inspired engineering.
   d. **Makerspace (Project-Based Learning):** Combines elements of traditional workshop practice with real-world project building and innovation, fostering design thinking and problem-solving.
   e. **Workshop Trades Modernized:** Generic workshop trades have been customized to be **branch-specific**, ensuring direct industry relevance and skill development.
   f. **Branch-Specific Core Engineering Courses:** Introduced in the first year to help students engage early with their chosen discipline and improve retention and academic interest.
   g. **Branch-Specific Science Course:** Designed to complement core engineering subjects with relevant scientific principles from the first year itself.
   h. **Structured Programming Methodology:** Introduced with **flexible language platforms** (C/Python/Java etc.) based on needs of the branch, enhancing adaptability and practical coding skills.
   i. **Object-Oriented Programming Methodology:** Added as a full course to strengthen software design thinking and align with industry practices.
   j. **End-Semester Practical Examination for Programming Courses:** Assesses hands-on competencies alongside theory, ensuring balanced evaluation

k. **Separate Heads of Passing for CA and ES:** Both Continuous Assessment (CA) and End Semester Examination (ESE) now have minimum marks for passing, ensuring consistent academic effort

l. **Unified Curriculum – Common Syllabus for All:** The earlier "Group-wise" (C-group/P-group) system is removed; all students now take the same courses in the same semester, reducing administrative complexity and ensuring equity.

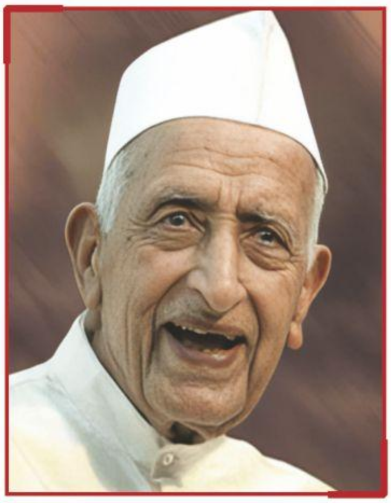
3. **Academic Advantages:**
   - Early exposure to core and specialized content increases engagement and program relevance.
   - Multidisciplinary inclusion (Biology, EVS, Makerspace) fosters 21st-century competencies
   - Emphasis on programming methodology and assessment balance improves computational thinking and hands-on skills
   - Value-based education and sports promote holistic development.

4. **Summary:**
   The SVU R-2025 First Year B. Tech. curriculum marks a strategic shift toward a more robust, inclusive, and application-oriented engineering education model. It ensures stronger alignment with national policies, global trends, and industry expectations with ease of administration

SOMAIYA VIDYAVIHAR UNIVERSITY

Somaiya TRUST

Our Vision



Padma Bhushan
Shri Karamshi Jethabhai Somaiya
Our Founder

Padma Bhushan Shri Karamshi Jethabhai Somaiya's first initiative in education was the founding of a school in 1942 in rural Maharashtra, to provide quality holistic education. It was founded on the belief that education is an important pillar of nation-building with the power to change lives and that it is the duty of the privileged to provide education to the needy.

He later founded the Girivanvasi Pragati Mandal, The K J Somaiya Medical Trust, the Girivanvasi Education Trust and sister institutions to make great citizens of India and the world. In the words of Swami Vivekananda, "We want that education by which character is formed, strength of mind is increased, the intellect expanded, and by which one can stand on one's own feet." We have now grown into a multi-disciplinary and multi-campus education institution with over 1500 faculty, and 39000+ students.

ज्ञानादेव तु कैवल्यम्।

KNOWLEDGE ALONE LIBERATES

Our motto is: ज्ञानादेव तु कैवल्यम्। Knowledge alone liberates. Liberates from poverty, from hunger. Also to liberate one from the attachments that bind us to small-mindedness. Knowledge also provides opportunity. To make the life lived more meaningful. In the service of one's family, one's community, one's समाज, country, and indeed the world. Bearing in mind that there is no religion other than the life lived in the service of humanity.

न मानुषात् परो धर्मः।

We will strive to provide access and opportunity to build a more inclusive society.

Our education in any subject will reflect its timeless fundamentals, its current context, and its applications. There is so much scientific discovery taking place, at the intersection of fields, in biology, computing, medicine, the social sciences and everywhere else. We will provide students and faculty with an environment to engage with this world, discover new truths, and make new applications to create and share knowledge.

Our education will also be experiential. With projects that are 'real' and those that complement the learning inside the classroom. Our students and faculty will be at the cutting edge of change, to incubate companies, to create NGOs, and pursue any field of their passion.

Our education will also be holistic. Sports and physical exercise must be a firm part of the curriculum. For students to develop a love for sports, for recreation, for health, for teamwork, and for competition. Our education will also instil an appreciation for art, culture, nature and biodiversity.

अभ्यासेन तु कौन्तेय वैराग्येण च गृह्यते।

In the Bhagavad Gita, Arjun asks Krishna how is one to control one's mind, which is as fleeting as the wind. Krishna responds that it can only be done through practice and discipline. अभ्यासेन तु कौन्तेय वैराग्येण च गृह्यते। We will strive to teach our students to learn to stay calm in our turbulent world. And our education will also include the ancient Indian tradition, its culture, its depth, and its knowledge. We must keep the connection with our mother tongue and our languages. Languages are storehouses of culture, and the loss of a language takes with it much learning, stored through it over the ages.

Finally, our education will help students lead a full life, and to fall in love with life. Our dream is to build a world-class research and teaching institution, that is global in the reach of its ideas, and universal in its service. Welcome to our community.

Our Mission

To nurture excellence and provide freedom of possibilities in education and research to foster a culture of creativity, innovation, leadership, responsible citizenship, service, and all-round growth.

- **Acronyms use:**

1. Acronyms for category of courses and syllabus template

Acronym	Description	Acronym	Description
BS	Basic Science Course	**CA**	Continuous Assessment (Theory Course)
ES	Engineering Science Course	**ESE**	End Semester Exam
HSS	Humanities and Social Sciences Course	**IA**	Internal Assessment
PC	Professional Core Course	**TH**	Theory
PE	Professional Elective course	**TUT**	Tutorial
OET	Open Elective - Technical	**MSE**	Mid- Semester Examination
OEM	Open Elective - Management	**Lab/Tut CA**	Continuous Assessment of Laboratory/Tutorial Course
LC	Laboratory Course	**PR/OR**	Practical/Oral Examination
PR	Project	**RM**	Research Methodology Course
VAC	Value-Added Course	**CO**	Course Outcome
VESC	Vocational Skill Enhancement Course	**PO**	Program Outcome
AEC	Ability Enhancement Course	**PSO**	Program Specific Outcome
CC	Co-curricular Course	**PEO**	Program Educational Objective

2. Type of Course

Acronym	Description
C	Core Course
E	Elective Course
O	Open Elective Technical
H	Open Elective - Humanities/ Management/ Swayam-NPTEL/ Coursera
P	Project
L	Laboratory Course
T	Tutorial

3. Ten Digit Course code e.g. 116U06C101

Acronym Serially as per code	Description
3	SVU-2025 Third Revision
16	College code
U	Alphabet code for type of program
06	Program/Department code
C	Type of course
1	Semester number (Semester I)
01	Course serial number

Knowledge and Attitude Profile (WK)
(As per NBA Criteria w. e. f. 2024-25)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues

WK9: Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Program Outcomes (POs) (As per NBA Criteria w. e. f. 2024-25)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problem

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyse and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Somaiya Vidyavihar University
K J Somaiya School of Engineering

FY B Tech (Common to All)
SVU R-2025 Version 3.0
SEM I

Teaching and Credit Scheme

Course Code		Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
SVU	KJSSE						
16U1001	316U06C101	BS	Applied Mathematics – 1	3 – 0 – 1	4	3 – 0 – 1	4
16U1002	316U06C102	BS	Engineering Physics	2 – 0 – 0	2	2 – 0 – 0	2
16U1003	316U06C103	BS	Engineering Chemistry	2 – 0 – 0	2	2 – 0 – 0	2
16U1004	316U06C104	ES	Basic Electrical Engineering	2 – 0 – 0	2	2 – 0 – 0	2
16U1005	316U06C105	ES	Engineering Drawing	2 – 0 – 1	3	2 – 0 – 1	3
16U1006	316U06C106	BS	Biology for Engineers	2 – 0 – 0	2	2 – 0 – 0	2
16U1007	316U06C107	ES	Structured Programming Methodology	2 – 2 – 0	4	2 – 1 – 0	3
16U1008	316U06L101	BS	Applied Science Laboratory - I	0 – 2 – 0	2	0 – 1 – 0	1
16U1009	316U06L102	ES	Basic Electrical Engineering Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
16U1010	316U06L103	ES	Engineering Drawing Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
16U1011	316U06L104	ES	Maker Space Laboratory – I	0 – 2 – 0	2	0 – 1 – 0	1
			SVU Basket Course*				
			Total	15 – 10 – 2	27	15 – 5 – 2	22

*****Students are required to complete one basket course in each of the first six semesters. By the end of Semester VI, they must have successfully completed one course from each of the six designated course baskets contributing to total of 12-18 credits.

Examination Scheme

Course Code		Course Category	Name of the Course	Lab/ Tut CA	CA		ESE		Total
SVU	KJSSE				IA	MSE	PR/OR Exam/ OST/ Project	Theory Exam	
16U1001	316U06C101	BS	Applied Mathematics – I	25	20	30	--	50	125
16U1002	316U06C102	BS	Engineering Physics	--	20	30	--	50	100
16U1003	316U06C103	BS	Engineering Chemistry	--	20	30	--	50	100
16U1004	316U06C104	ES	Basic Electrical Engineering	--	20	30	--	50	100
16U1005	316U06C105	ES	Engineering Drawing	--	20	30	--	50	100
16U1006	316U06C106	BS	Biology for Engineers	--	50	--	--	--	50
16U1007	316U06C107	ES	Structured Programming Methodology	50	--	--	50	--	100
16U1008	316U06L101	BS	Applied Science Laboratory - I	50	--	--	--	--	50
16U1009	316U06L102	ES	Basic Electrical Engineering Laboratory	50	--	--	--	--	50
16U1010	316U06L103	ES	Engineering Drawing Laboratory	50	--	--	--	--	50
16U1011	316U06L104	ES	Maker Space Laboratory – I	50	--	--	--	--	50
			SVU Basket Course**						
			Total	275	150	150	50	250	875

**Marks obtained in basket courses will be scaled on a Evaluation of basket courses will be based on scaled up/down marks out of 100Students are required to complete one basket course in each of the first six semesters. By the end of Semester VI, they must have successfully completed one course from each of the six designated course baskets contributing to total of 12-18 credits.

Mandatory internship: It is mandatory for every student to complete **10 weeks of internship** spanning across the four years of B. Tech Program over and above the academic credits. Students can take up internships in community services / socially relevant projects (optional and limited to 4 weeks) and in the technical domain (minimum 6 weeks or more). Students will be awarded an internship completion certificate along with their graduation.

FY B Tech (Common to All)
SVU R-2025 Version 3.0
SEM II

Teaching and Credit Scheme

Course Code SVU	Course Code KJSSE	Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
16U1012	316U06C201	BS	Applied Mathematics – II	3 – 0 – 1	4	3 – 0 – 1	4
16U1022 16U1023 16U1018	316U06E211 316U06E212 316U06E221	BS BS ES	Applied Science for Computer & Allied Programs/ Applied Science for Electronics & Allied Programs/ Engineering Mechanics †	3 – 0 – 0	3	3 – 0 – 0	3
16U1019	316U06C203	ES	Digital Logic Design	3 – 0 – 0	3	3 – 0 – 0	3
16U1013	316U06C204	HSS	Environmental Science	2 – 0 – 0	2	2 – 0 – 0	2
16U1014	316U06C205	ES	Object-Oriented Programming Methodology	2 – 2 – 0	4	2 – 1 – 0	3
16U1015	316U06T201	HSS	Presentation and Communication Skills	1 – 0 – 1	2	1 – 0 – 1	2
16U1016 16U1021	316U06L201 316U06L221	BS ES	Applied Science Laboratory - II/ Engineering Mechanics Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
16U1020	316U06L202	ES	Digital Logic Design Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
16U1017	316U06L203	ES	Maker Space Laboratory – II	0 – 2 – 0	2+1#	0 – 1 – 0	1
	--		SVU Basket Course				
			Total	14 – 8 – 2	24+1#	14 – 4 – 2	20

† Only for MECH/RAI students (in place of Applied Science)
#Additional 1 hr/division/week is assigned for project review of PBL component (only for timetable slot. No separate course credit is given)

Examination Scheme

Course Code SVU	Course Code KJSSE	Course Category	Name of the Course	Lab/ Tut CA	CA IA	CA MSE	ESE PR/OR Exam/ OST/ Project	ESE Theory Exam	Total
16U1012	316U06C201	BS	Applied Mathematics – II	25	20	30	--	50	125
16U1022 16U1023 16U1018	316U06E211 316U06E212 316U06E221	BS BS ES	Applied Science for Computer & Allied Programs/ Applied Science for Electronics & Allied Programs/ Engineering Mechanics †	--	20	30	--	50	100
16U1019	316U06C203	ES	Digital Logic Design	--	20	30	--	50	100
16U1013	316U06C204	HSS	Environmental Science	--	50	--	--	--	50
16U1014	316U06C205	ES	Object Oriented Programming Methodology	50	--	--	50	--	100
16U1015	316U06T201	HSS	Presentation and Communication Skills	50	--	--	--	--	50
16U1016 16U1021	316U06L201 316U06L221	BS ES	Applied Science Laboratory - II/ Engineering Mechanics Laboratory	50	--	--	--	--	50
16U1020	316U06L202	ES	Digital Logic Design Laboratory	50	--	--	--	--	50
16U1017	316U06L203	ES	Maker Space Laboratory – II	50	--	--	50$	--	100
	—		SVU Basket Course						
			Total	275	110	90	100	150	725

$Project evaluation applicable only for PBL component of Makerspace Laboratory

Detailed Syllabus (Semester I)

Course Code	Course Title			
316U06C101	Applied Mathematics – I			
Teaching Scheme (hrs.)	TH	PRACT	TUT	Total
	03	--	01	04
Credits Assigned	03	--	01	04

Examination Scheme	Marks					
	CA		ESE (Theory/On-Screen)	PR/OR Exam	LAB/TUT CA	Total
	IA	MSE				
	20	30	50	--	25	125

Course prerequisites:
Basics of matrices, inverse and adjoint, differentiation techniques, differential equations, complex numbers

Course Objectives:
The course introduces concept of rank of matrix, solving system of linear equations is explained in detail with some applications. The course introduces the concept of partial differentiation and its applications to find extreme values of a function and Jacobian. The course communicates the methods of solving linear differential equations. The objective of the course is to impart knowledge of De-Moivre's theorem, hyperbolic functions and logarithm of complex numbers

Course Outcomes
At the end of the successful completion of the course, the student will be able to
CO1: Apply the concept of rank of a matrix and numerical methods to solve system of linear equations
CO2: Find partial derivatives of multivariable functions, apply the concept of partial differentiation to find maxima and minima of 2 variable functions
CO3: Apply Euler's theorem to prove results related to Homogeneous functions.
CO4: Identify and solve different types of ordinary differential equations using various methods.
CO5: Solve problems involving different forms and properties of complex numbers, hyperbolic functions and logarithm of complex numbers

Module No.	Unit No.	Details	Hrs.	CO
1		**Rank of Matrix and System of Equations**	08	CO1
	1.1	Types of matrices: Hermitian, Skew-Hermitian, Unitary and Orthogonal matrix, Encryption and Decryption using orthogonal matrix		
	1.2	Rank of a matrix using row echelon forms, reduction to normal form		
	1.3	System of homogeneous and non-homogeneous equations, their consistency and solutions		
	1.4	Linearly dependent and independent vectors		
	1.5	Introduction of transitioning from Linear Systems to Non-Linear System of Equations, Solution of system of linear algebraic equations by (a) Gauss-Seidal method (b) Jacobi iteration method		
	#	**Self-learning topics**: Symmetric, Skew-symmetric matrices and properties, Properties of adjoint and inverse of a matrix, Properties of orthogonal and unitary matrix, Solving System of linear equation using any mathematical software		
2		**Partial Differentiation and Application**	09	CO2
	2.1	Functions of several variables, Partial derivatives of first and higher order (definition using limits and simple problems)		
	2.2	Differentiation of composite functions		
	2.3	Maxima and minima of a function of two independent variables		
	2.4	Introduction of Jacobian of two and three independent variables (simple problems)		
3		**Homogeneous Functions**	04	CO3
	3.1	Euler's theorem on homogeneous functions with two and three independent variables (statement only) and problems		
	3.2	Deductions(Corollaries) from Euler's theorem (statements only) and problems		
4		**Linear Differential Equations of First and Higher Order**	12	CO4
	4.1	Differential Equation of first order and first degree- Exact differential equations, Equations reducible to exact equations by integrating factors.		
	4.2	Linear differential equations (Review), Equation reducible to linear form. Applications of Differential Equation of first order and first degree		
	4.3	Higher order Linear Differential Equation with constant coefficients: Complimentary function, particular integrals of differential equation of the type $f(D)y = X$, where X is e^{ax}, $sin\,(ax + b), cos\,(ax + b\,), x^n$, $e^{ax}V$		
	4.4	Method of variation of parameters		
	#	**Self-learning topic**: Bernoulli's equation. Equation reducible to Bernoulli's equation. Cauchy's homogeneous linear Differential Equation, Applications of Higher Order Differential Equations		
5		**Complex Numbers, Hyperbolic Functions and Logarithm of Complex Number**	12	CO5
	5.1	Statement of De Moivre's theorem and related examples		
	5.2	Powers and roots of complex numbers		
	5.3	Circular functions and hyperbolic functions of complex number		
	5.4	Inverse circular and inverse hyperbolic functions		
	5.5	Logarithm of complex numbers		
	5.6	Separation of real and imaginary parts of a function		

	#	**Self-learning topics**: Expansion of $sinn\theta$, $cosn\theta$ in terms of sine and cosine of multiples of angle θ and expansion of $sinn\theta$, $cosn\theta$ in powers of $sin\theta$, $cos\theta$		
		Total	**45**	

Students should prepare all self-learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in Tutorials.

Recommended Books:

Sr. No.	**Name/s of Author/s**	**Title of Book**	**Name of Publisher with country**	**Edition and Year of Publication**
1.	B. S. Grewal	Higher Engineering Mathematics	Khanna Publications, India	43rd Edition 2014
2.	Shanti Narayan	A text book of Matrices	S. Chand, India	10th Edition 2004
3.	Erwin Kreyszig	Advanced Engineering Mathematics	Wiley Eastern Limited, India	10th Edition 2015
4.	Ramana B.V.	Higher Engineering Mathematics	Tata Mcgraw Hill New Delhi, India	34th Edition (reprint) 2019
5.	Glyn James	Advanced Modern Engineering Mathematic	Pearson Publication India	4th Edition 2010
6.	M. D. Raisinghania	Ordinary and Partial Differential Equations	S. Chand, India	18th Edition 2013

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<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06C102</td><td colspan="6">Engineering Physics</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>02</td><td colspan="2">--</td><td colspan="2">--</td><td>02</td></tr>
<tr><td>Credits Assigned</td><td>02</td><td colspan="2">--</td><td colspan="2">--</td><td>02</td></tr>
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<table>
<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td>100</td></tr>
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Course prerequisites:

Physics: Metric units and conversions, basic concepts and laws of optics, electricity and magnetism, electrical properties of conductors and semiconductors, particle properties of radiation

Mathematics: Basics of Differentiation and integration, differential equations, vector operations, trigonometric identities, logarithms

Course Objectives:

This course is designed to establish strong foundations of Engineering Sciences by using a problem-solving approach to learn fundamental physical concepts and mathematical formulation of a variety of real-life applications. The course covers areas of both, pure and applied Physics such as wave optics, laser and fibre optics, electromagnetism. This course promotes hands-on and inquiry-based learning and aligns with multidisciplinary frameworks

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1: Explain different optical phenomena using concepts of wave optics

CO2: Develop conceptual understanding of the working of lasers and optical fibers

CO3: Understand the basic principle of quantum mechanics

CO4: Analyze semiconductor parameters and solve problems using mathematical foundations of electromagnetism

Module No.	Unit No.	Details	Hrs.	CO
1		**Wave Optics**	08	CO1
	1.1	Interference: Introduction, methods of interference, Derivation of interference in thin parallel films, Theoretical discussion of a wedge shaped film and Newton's Ring		
	1.2	Diffraction: Introduction, Theoretical discussion of diffraction grating and Resolving power		
2		**Photonics**	07	CO2
	2.1	**Lasers**: Interaction of radiation with matter, Basic principles and components required for lasing, Relation between Einstein's coefficient, Threshold condition of laser		
	2.2	**Fiber optics**: Concept of total internal reflection, Types of optical fiber, Derivation of Numerical aperture for step index fiber, Important parameters related an optical fiber, losses in optical fibers		
3		**Quantum Mechanics – I**	07	CO3
	3.1	De-Broglie hypothesis and its various forms		
	3.2	Heisenberg's uncertainty principle, its implications and application		
	3.3	Wavefunction and its probabilistic interpretation, Condition of Normalization		
4		**Basics of Semiconductors and Electromagnetism**	08	CO4
	4.1	Intrinsic and Extrinsic semiconductors, Mobility, Conductivity in semiconductors		
	4.2	Drift and Diffusion current densities		
	4.3	Fermi- Dirac distribution function, Position of Fermi Level in Intrinsic and Extrinsic Semiconductors		
	4.4	Del operator, Gradient, divergence, curl, Fundamental theorems and their physical interpretation		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	M N Avadhanulu, P G Kshirsagar	A Textbook of Engineering Physics	S Chand	11th Edition, 2018
2	Gaur, Gupta	Engineering Physics	Dhanpat Rai, India	8 Edition, 2018
3	Ajay Ghatak	Optics	McGraw Hill India	6th Edition, 2017
4	S. M. Sze	Physics of Semiconductors	Wiley	7th Edition, 2017
5	David Griffiths	Introduction to Electrodynamics	PHI	5th Edition, 2015

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<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06C103</td><td colspan="6">Engineering Chemistry</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td>PRACT</td><td colspan="3">TUT</td><td>Total</td></tr>
<tr><td>02</td><td>--</td><td colspan="3">--</td><td>02</td></tr>
<tr><td>Credits Assigned</td><td>02</td><td>--</td><td colspan="3">--</td><td>02</td></tr>
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<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td>100</td></tr>
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Course prerequisites:

A foundational understanding of chemical principles, atomic structure, periodic classification, chemical bonding, and states of matter, basic organic and inorganic chemistry, and simple numerical problem-solving skills in chemistry is essential. This background will enable students to grasp the advanced concepts related to materials science, electrochemistry, corrosion, polymers, nanomaterials, and environmental chemistry encountered in the course.

Course Objectives:

The objective of course is to develop a strong conceptual foundation in fundamental chemical principles relevant to engineering applications and to appreciate the basic concepts of chemistry behind development of futuristic materials and their applications in engineering and technology. The course objective is to understand chemical processes involved in development of sustainable energy sources.

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1: Understand and apply key parameters used to assess water pollution such as BOD and COD and explore modern techniques for the treatment of industrial wastewater.

CO2: Understand and apply the principles and practices of Green Chemistry with a focus on designing environmentally benign chemical processes.

CO3: Apply the knowledge of synthetic organic chemistry for justifying mechanism of chemical reactions and appreciate its role for industrial development.

CO4: Apply electrochemical principles to analyze corrosion behavior and design practical protection strategies such as coatings, inhibitors, and cathodic/anodic methods for engineering systems

Module No.	Unit No.	Details	Hrs.	CO
1	**Industrial Applications of Water and its treatment methods**		**08**	**CO1**
	1.1	Introduction, Types of Hardness, Equivalence of CaCO3, Experimental determination of hardness.		
	1.2	Emerging Technology for Sustainable Water Treatment: Lime soda method Zeolite method, Ion Exchange process.		
	1.3	Methods to determine extent of water pollution, BOD, COD, Treatment of industrial wastewater.		
2	**Green Chemistry for Sustainable Engineering**		**08**	**CO2**
	2.1	Principles and Concept of Green Chemistry: Definition and need for green chemistry in modern chemical industries, Twelve principles of green chemistry (Anastas and Warner), significance of each principle with respect to sustainability goals.		
	2.2	Green solvents: water, supercritical CO_2, ionic liquids, Comparison of traditional vs green chemical processes, Atom economy, waste minimization, and environmental metrics		
	2.3	Green Synthesis and Applications: Microwave- and ultrasound-assisted green synthesis, Case studies of industrial green chemistry practices		
3	**Synthetic Organic reactions**		**07**	**CO3**
	3.1	Name reactions: i) Aldol Condensation, ii) Baeyer –Villiger oxidation, iii) Dakin Reaction, iv) Haloform reaction, v) Sharpless Epoxidation, 6) Wurtz synthesis		
	3.2	Rearrangement Reaction: 1) Benzil- Benzilic Acid Rearrangement, 2) Pinacole-Pinacolone Rearrangement 3) Bayer-Villiger Rearrangement 4) Fries Rearrangement		
4	**Electrochemistry & Corrosion Fundamentals**		**07**	**CO4**
	4.1	Nernst equation, electrode potentials, cell constructions, Definition and significance of corrosion in engineering: Types of Corrosion		
	4.2	Factors affecting rate of corrosion, Methods to control corrosion: Material Selection and Purity Improvement, Design Considerations to Minimize Corrosion, Use of Protective Coatings and Paints, Cathodic Protection, Anodic Protection, Use of Corrosion Inhibitors, Environmental Control, Alloying to Improve Corrosion Resistance		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Engineering Chemistry	K. SESHA Maheswaramma, Mridula Chugh	Pearson	10th Edition, 2016
2.	A textbook of Engineering Chemistry	Dr. S.S.Dara, Dr. S.S. Umare	S. Chand	9th Edition, 2015
3.	Organic reactions Mechanisms	V.K. Ahluwalia, Rakesh Kumar Parashar	Narosa	3rd Edition, 2015
4.	Green Chemistry: Theory and Practice	Paul T. Anastas , John Warner	Oxford University Press	2nd Edition, 2000
5.	A Guidebook to Mechanism in Organic Chemistry	Peter Sykes	Pearson Education	4th Edition, 2003

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<tr><td>316U06C104</td><td colspan="6">Basic Electrical Engineering</td></tr>
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<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td>100</td></tr>
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Course prerequisites:
Knowledge of Basic Electrical parameters: Resistance, Inductance, Capacitance, Frequency, Voltage, Current and Power and Energy, basic laws of magnetism

Course Objectives:
It is difficult to imagine life without electricity. Electricity plays a major role in the working of all minor and major devices used in our day to day life. In this course students acquire fundamental knowledge to understand the electrical systems

Course Outcomes
At the end of the successful completion of the course, the student will be able to
CO1: Analyze resistive DC networks using various network theorems
CO2: Analyze single phase AC circuits
CO3: Analyze three phase AC circuits
CO4: Understand principles and working of Transformer
CO5: Describe principles and working of DC and AC motors

Module No.	Unit No.	Details	Hrs.	CO
1		**DC Circuits**	**10**	**CO1**
	1.1	Concept of dependent and independent sources, ideal and practical voltage and current sources, Kirchhoff's Laws, source transformation and network terminology.		
	1.2	Resistive network simplification, Series, parallel connection and Star-Delta transformations		
	1.3	Mesh and nodal analysis, concept of super mesh and super node (Analysis only with independent sources)		
	1.4	Superposition theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem (Analysis only with independent sources)		
2		**Single Phase AC Circuits**	**10**	**CO2**
	2.1	Generation of alternating voltage, average value, RMS value, form factor, crest factor, phasor representation in rectangular and polar form.		
	2.2	Steady state behaviour of single phase AC circuits with pure R, L, and C, concept of inductive and capacitive reactance, phasor diagram of impedance, phase relationship in voltage and current.		
	2.3	RL, RC and RLC series and parallel circuits, concept of impedance and admittance, power triangle, power factor, active, reactive and apparent power, concept of power factor improvement.		
	2.4	Series and parallel resonance, Q-factor and bandwidth		
3		**Three Phase AC circuits**	**05**	**CO3**
	3.1	Three-phase balanced circuits, voltage and current relations in star and delta connections.		
	3.2	Measurement of power in 3-phase system using two wattmeter method		
4		**Introduction to Electrical Machines**	**05**	**CO4**
	4.1	**Single phase transformer :** Construction and principle of working, emf equation of a transformer, losses in transformer, Equivalent circuit of Ideal and practical transformer, and efficiency of transformer, (No numerical expected)		
	4.2	**DC motors** : Construction and working principle of DC motors such as series, shunt and compound, torque-speed characteristics, selection criteria and applications (no derivations and numerical expected)		
	4.3	**Single phase induction motor**: Construction, working principle, double field revolving theory, applications (no derivations and numerical expected)		
	#	**Self-Learning:** BLDC Motors, PMSM and Stepper Motors, SRM Motor, Servo Motor, Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthling, Lamps: fluorescent, CFL, LED, Electrical measuring instruments principle and applications-energy meter, megger, tong tester.		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	B. L. Thereja	Electrical Technology Vol-1 and Vol-II	S. Chand & Publications, India	25^{th} Edition 2014
2.	Mittle and Mittle	Basic Electrical Engineering	Tata McGraw Hill, India	2^{nd} edition (New) 2001
3.	Singh Ravish R	Basic Electrical Engineering	S. Chand	1^{st} Edition, 2023
4.	B.R. Patil	Basic Electrical Engineering	Oxford University Press	2^{nd} Edition, 2022

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<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td colspan="2">100</td></tr>
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Course prerequisites:
Knowledge of various geometric constructions, Basics of trigonometry

Course Objectives:
The objective of course is to make students familiarize with the conventions and standards of engineering drawing along with the principles of projections applied to points, lines and planes, apply the principles of orthographic projections to draw elevation, plan, side view, and isometric views etc., apply the principles of orthographic projections to draw various views of regular solids and apply the fundamentals of sectional view to the regular solids and develop lateral surfaces of solids

Course Outcomes
At the end of the successful completion of the course, the student will be able to
CO1: Visualize and draw orthographic projection and sectional views of any 3D object.
CO2: Visualize and draw projection of lines and planes
CO3: Draw projection of regular solids
CO4: Draw sectional views and lateral development of regular solids
CO5: Visualize and draw isometric drawing

Module No.	Unit No.	Details	Hrs.	CO
1	**Orthographic Projection**		**06**	**CO1**
	1.1	Introduction to Engineering Drawing, Standard sizes of drawing sheets, Types of Lines, Dimensioning, Scales, Drawing Pencils etc.		
	1.2	Orthographic projections of simple machine parts by first angle method as recommended by Indian standards		
	1.3	Sectional views of simple machine parts (full section ONLY)		
2	**Projection of Points, Lines And Planes**		**08**	**CO2**
	2.1	Projection of points, Projection of lines inclined to both the reference planes. (Line in 1st quadrant ONLY)		
	2.2	Projection of Planes: Triangular, Square, Rectangular, Pentagonal, Hexagonal and circular planes inclined to one reference plane only and perpendicular to other		
3	**Projection of Solids**		**06**	**CO3**
	3.1	Introduction to Projection of Solids, Classification of Solids		
	3.2	Projection of right regular solids (prism, pyramid, cylinder, and cone) inclined to one reference plane only		
4	**Section and Development of Solids**		**06**	**CO4**
	4.1	Projection of sectional views of solids (prism, pyramid, cylinder, and cone) cut by the plane perpendicular to one and inclined to other reference plane only		
	4.2	Lateral surface development of solids (prism, pyramid, cylinder, and cone) cut by the section plane inclined to one reference plane only		
5	**Isometric View**		**04**	**CO5**
	5.1	Introduction to Isometric view/drawing, isometric projection		
	5.2	Construction of isometric drawing of simple machine parts		
	#	**Self-Learning**: 3D modelling using AUTOCAD		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	N.D. Bhatt	Engineering Drawing (Plane and solid geometry)	Charotar Publishing House Pvt. Ltd. India	52nd Revised and enlarged Edition
2.	N.D. Bhatt V.M. Panchal	Machine Drawing	Charotar Publishing House Pvt. Ltd. India	48th Edition
3.	P. S. Gill	Engineering Graphics and Drafting	S.K. Kataria & Sons, India	11th Edition
4.	P.J. Shah	Engineering Graphics	S. Chand Publications, India	Revised Edition
5.	Dhananjay Jolhe	Engineering Drawing	Tata McGraw Hill, India	Revised Edition

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<tr><td>316U06C106</td><td colspan="6">Biology for Engineers</td></tr>
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<tr><td>IA</td><td>MSE</td></tr>
<tr><td>50</td><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td></tr>
</table>

Course prerequisites:
Basics of biological systems

Course Objectives:
Biology for Engineers is an interdisciplinary course designed for the students of various engineering streams to appreciate the link between biological Science and engineering

Course Outcomes
At the end of the successful completion of the course, the student will be able to
CO1: Describe the role of biomolecules and enzymes and evaluate their industrial applications
CO2: Explain the structure, function, and molecular biology of cells and apply basic bioinformatics tools
CO3: Illustrate fermentation processes, bioprocess equipment, and their industrial relevance
CO4: Analyze the design, principles, and applications of biosensors and biomedical instruments
CO5: Understand system biology and analysis
CO6: Apply biomechanical concepts and evaluate materials based on biocompatibility for biomedical applications.

Module No.	Unit No.	Details	Hrs.	CO
1	**Biomolecules and Enzymes**		05	CO1
	1.1	Lipids, carbohydrates, amino acids and proteins, nucleic acids		
	1.2	Enzymes and Industrial applications: Enzymes, endo-enzymes and exo-enzymes, enzyme action, Types of enzymes, Cofactors , Enzyme Kinetics, Industrial Applications of Enzymes		
2	**Basic Cell Biology and bioinformatics**		06	CO2
	2.1	Basic definition of a cell, prokaryotic cell , eukaryotic cell, cell cycle and cell division, m-phase, meiosis, cell differentiation		
	2.2	Nucleotides, DNA, RNA, tRNA, mRNA		
	2.3	DNA replication, transcription, translation		
	2.4	Introduction to bioinformatics, applications of bioinformatics		
3	**Bioprocess Engineering**		05	CO3
	3.1	Design of Fermentors - Types, Components, sensors		
	3.2	Upstream and downstream process- Fermentation kinetics, monod kinetics, Thermal kinetics		
	3.3	Case studies of fermentation of products and processes – Enzyme, acids, antibiotics		
4	**Biosensors and Bioelectronics**		04	CO4
	4.1	Design and Applications of Biosensors and Biofuels		
	4.2	Voltage Sensors, Optical Sensors, Displacement/Pressure Sensors and Accelerometers, Chemical Sensors, Acoustic Sensors		
	4.3	Heart, Working of Heart, Heartbeat monitoring, EEG, ECG, Sonography, MRI, Tomography		
5	**Systems Biology**		05	CO5
	5.1	Systems biology of metabolic pathways		
	5.2	Systems analysis and modelling in medicine, agriculture and climate change		
	5.3	Systems dynamics, control and optimization		
	Biomechanics and Biocompatible materials		05	CO6
	6.1	Introduction of biomechanics; anatomical concepts in biomechanics		
	6.2	Musculoskeletal biomechanics and Cardiovascular mechanics: musculoskeletal geometry, muscle structure and force generation, motion tracking techniques, cardiovascular physiology, Blood Flow Models		
	6.3	Case studies on applications of biomechanics on bones, joints, muscles, tissues etc.		
	6.4	Physico-chemical properties of biomaterials: mechanical (elasticity, yield stress, ductility, toughness, strength, fatigue, hardness, wear resistance), tribological (friction, wear, lubricity), morphology and texture, physical (electrical, optical, magnetic, thermal), chemical and biological properties.		
	6.5	Technologies of biomaterials processing, as implants and medical devices; improvement of materials biocompatibility by plasma processing.		
	6.6	Tissue engineering		
		Total	30	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	G. K. Suraishkumar	Biology for Engineers	Oxford University Press	2nd Edition, 2019
2.	Wiley Editorial	Biology for Engineers	Wiley	3rd Edition, 2018
3.	Andrew G. Webb	Principles of Biomedical Instrumentation,	Cambridge Texts in Biomedical Engineering.	9th Edition, 2018
4.	Campbell, N. A.	Biology: A global approach	Pearson Education Ltd.	11th Edition
5.	Jin Xiong	Essential Bioinformatics	Cambridge University Press	Edition 2007
6.	S. Ignacimuthu	Basic Bioinformatics	Narosa Publishing House	2md Edition 2013
7.	Cees Oomens, Marcel Brekelmans, Frank Baaijens	Biomechanics: concepts and computation	Cambridge texts in Biomedical Engineering	2nd Edition
8.	Peter F. Stanbury, Allan Whitaker, Stephen J. Hall	Principles of Fermentation Technology	Butterworth-Heinemann (Elsevier)	2nd Edition 1995

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<tr><td>316U06C107</td><td colspan="7">Structured Programming Methodology</td></tr>
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*The End Semester Examination (ESE) will be conducted in an onscreen format and may include a combination of theoretical questions, practical coding tasks, and/or an oral/viva component

Course prerequisites:

Students taking this course should have a basic understanding of computer operations, including familiarity with operating systems and file management. They are expected to possess fundamental logical reasoning skills and the ability to approach problems methodically. A basic grasp of mathematics, particularly arithmetic and simple algebra

Course Objectives:

The objective of course is to understand structured programming concepts and basic problem-solving techniques using algorithms, flowcharts, and pseudocode, implement decision-making and looping constructs to control program flow effectively, use arrays and strings for data storage and manipulation and apply functions, structures, and pointers to create modular and efficient programs.

Course Outcomes

At the end of the successful completion of the course, the student will be able to understand

CO1: Formulate a problem statement and develop the logic (algorithm/flowchart) for its solution
CO2: Demonstrate the use of control structures
CO3: Apply the concepts of arrays and strings
CO4: Design modular programs using functions and the use of structure

Module No.	Unit No.	Details	Hrs.	CO
1		**Introduction to Structured Programming methodology**	**05**	**CO1**
	1.1	Problem solving skill development: Problem Definition, fundamentals of algorithms and flowcharts, Program Design, Pseudocode.		
	1.2	Structured Programming		
	1.3	Program execution process, Systems Development Life Cycle		
	1.4	Understanding concept and importance of header file/ package/namespaces Data & Operators: Data Types, Identifier, Constants and Variables		
	1.5	Types of Operators, Expressions and Evaluation of Expressions, Operator Precedence and Associativity, Type Conversions		
2		**Program Control Functions**	**06**	**CO2**
	2.1	Decision Making and Branching Control Structures: Two Way Selection, Multiway Selection		
	2.2	Looping Control Structures, Flag Concept, Counting Loops		
	2.3	Documentation and Making Source Code Readable		
3		**Introduction to Arrays**	**07**	**CO3**
	3.1	Arrays: Introduction to One Dimensional Arrays, Multidimensional Arrays, Declaration and Initialization of Arrays, Reading and Displaying arrays		
	3.2	Character Arrays and Strings: Introduction, Declaring and Initializing String Variables, Reading Character and Writing Character, Reading and Writing Strings, various operation on strings, Implementation of string handling operations (from scratch)		
4		**User defined function and Structures**	**12**	**CO4**
	4.1	User Defined Functions: Need, Function Declaration and Definition, Return Values, Function Calls, Passing Arguments to a Function by Value, Recursive functions, String Handling Functions (inbuilt)		
	4.2	Structures and Unions: Introduction, Declaring and defining Structure, Structure Initialization, Accessing and Displaying Structure Members, Array of Structures		
	4.3	Introduction to pointers: Pointer declaration and initialization, Pointer addition and subtraction, evaluating pointer expressions Pointers and Functions: Pass by Reference, Returning pointers from functions		
	#	**Self-Learning**: Introduction to Unions, Structure Vs Unions, File Handling		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Kenneth Leroy Busbee	Programming Fundamentals - A Modular Structured Approach using C++	Rice University, Houston, Texas	2013
2	Hassan Afyouni Behrouz A. Forouzan	Computer Science: A Structured Programming Approach Using C	Cengage India Private Limited	4^{th} Edition 2023
3	Behrouz A. Forouzan Richard F. Gilberg	Computer Science: A Structured Approach Using C++	Cengage India Private Limited	2nd Edition 2012
4	E. Balagurusamy	Programming in ANSI C	McGraw-Hill Education, India	8^{th} Edition, 2019
5	Pradeep Dey and Manas Ghosh	Structured Programming Approach	Oxford University Press, India	1^{st} Edition, 2016
6	https://press.rebus.community/programmingfundamentals/chapter/structured-programming/			
7	https://tfetimes.com/wp-content/uploads/2015/04/structured-programming-with-c-plus-plus.pdf			
8	https://open.umn.edu/opentextbooks/textbooks/144			
9	NPTEL (C)	https://onlinecourses.nptel.ac.in/noc22_cs40/preview		
10	NPTEL (C++)	https://onlinecourses.nptel.ac.in/noc21_cs02/preview		

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<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
</table>

Note: List of experiments/tutorials, scheme for continuous assessment and rubrics will be shared by course coordinators from time to time at the beginning of every semester

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06L102</td><td colspan="6">Basic Electrical Engineering Laboratory</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>--</td><td colspan="2">02</td><td colspan="2">--</td><td>02</td></tr>
<tr><td>Credits Assigned</td><td>--</td><td colspan="2">01</td><td colspan="2">--</td><td>01</td></tr>
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<table>
<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
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<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
</table>

Note: List of experiments/tutorials, scheme for continuous assessment and rubrics will be shared by course coordinators from time to time at the beginning of every semester

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06L103</td><td colspan="6">Engineering Drawing Laboratory</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td>TUT</td><td colspan="2">Total</td></tr>
<tr><td>--</td><td colspan="2">02</td><td>--</td><td colspan="2">02</td></tr>
<tr><td>Credits Assigned</td><td>--</td><td colspan="2">01</td><td>--</td><td colspan="2">01</td></tr>
<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
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<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
</table>

Note: List of experiments/tutorials, scheme for continuous assessment and rubrics will be shared by course coordinators from time to time at the beginning of every semester

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06L104</td><td colspan="6">Maker Space Laboratory – I</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td colspan="2">TH</td><td colspan="2">PRACT</td><td>TUT</td><td>Total</td></tr>
<tr><td colspan="2">--</td><td colspan="2">02</td><td>--</td><td>02</td></tr>
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<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
</table>

Course prerequisites:
--

Course Objectives:
The main objective of the Maker Space Laboratory is to provide all engineering students with theoretical and practical knowledge of the manufacturing environment. This course is the foundation of the real industrial environment, and it helps students develop and improve relevant technical hand skills. It teaches the fundamentals of various hand tools, power tools, machine tools, and their applications in various areas of manufacturing. The Design Thinking Laboratory experiences would help in developing an understanding of the complexity of real world problems, as well as the time and skill requirements for finding solutions to the same

Note:
1. Each student must complete three shops during semester I. The shops compulsory for the specified branch are mentioned in the bracket
2. A student team will have four or five members. Each team will have a maximum of 10 hours to complete their assigned task in each shop
3. Assessment:
   i. Continuous assessment will be done throughout the semester for the allotted shops
   ii. Quality of finished product

Shop	Department/Program
Precision Sheet Metal Working Shop	Compulsory for All Programs
Project Based Learning (PBL)	Compulsory for All Programs
Computer Hardware and Assembly Shop	Computer Engineering, Information Technology
Printed Circuit Board (PCB) Shop	Electronics Engineering, Electronics and Telecommunication Engineering
Woodcraft and Fabrication Shop*	Mechanical Engineering
Fabrication Process Shop*	Mechanical Engineering

*Any one of woodcraft and fabrication OR Fabrication processes shop

Detailed Syllabus
(Semester II)

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06C201</td><td colspan="6">Applied Mathematics – II</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>03</td><td colspan="2">--</td><td colspan="2">01</td><td>04</td></tr>
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<table>
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<tr><td>IA</td><td>MSE</td></tr>
<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>25</td><td>125</td></tr>
</table>

Course prerequisites:
Rank of Matrix, system of Equations, Basics of Integration and Differentiation, Knowledge of standard curves

Course Objectives:
The objective of the course is to impart knowledge of Eigenvalues and Eigenvectors of a matrix, concept of diagonalization, principal component analysis and singular value decomposition. The course introduces the concept of successive differentiation and helps students to find a series of some standard functions. The course communicates various techniques to solve improper integrals. The concept of multiple integration is introduced and applications to find Area and Volume are discussed.

Course Outcomes
At the end of the successful completion of the course, the student will be able to

- **CO1.** Apply the concept of eigenvalues, eigenvectors of a matrix to diagonalization of a matrix, principal component analysis, singular value decomposition, Cayley-Hamilton theorem, and functions of square matrices
- **CO2.** Solve problems involving successive derivatives of real variable functions. Expand a function as an infinite series using Taylor’s and Maclaurin’s series
- **CO3.** Apply concept of Beta – Gamma function and DUIS to solve improper integrals
- **CO4.** Find length of a curve using Cartesian, Polar and Parametric equations of curves
- **CO5.** Evaluate multiple integrals and use it to find Area and Volume

Module No.	Unit No.	Details	Hrs.	CO
1	**Eigenvalues and Eigenvectors**		**12**	**CO1**
	1.1	Characteristic equation, Cayley-Hamilton theorem, Examples based on verification and application of Cayley-Hamilton theorem		
	1.2	Eigenvalues and Eigenvectors, Properties of Eigenvalues and Eigenvectors		
	1.3	Similarity of matrices, Diagonalization of a matrix		
	1.4	Functions of a square matrix		
	1.5	Application of Eigenvalues and Eigenvectors: Principal Component Analysis, Singular Value Decomposition		
	#	**Self-learning topics**: Derogatory and Non-Derogatory Matrices		
2	**Successive Differentiation, Expansion of Functions**		**6**	**CO2**
	2.1	Successive differentiation: n^{th} derivative of standard functions, Leibnitz's Theorem.		
	2.2	Expansion of Functions: Taylor's Theorem, Taylor's series and Maclaurin's series		
	#	**Self-learning topic**s: Expansion of hyperbolic and logarithmic functions, Indeterminate forms, L'Hospital Rule, expansion using standard series		
3	**Integration : Review and Some New Techniques**		**8**	**CO3**
	3.1	Beta and Gamma functions with properties.		
	3.2	Differentiation under integral sign with constant limits of integration.		
	#	**Self-learning topic**: Differentiation under integral sign with variable limits of integration		
4	**Rectification**		**4**	CO4
	4.1	Rectification of plane curves in Cartesian form.		
	4.2	Rectification in parametric and polar forms.		
5	**Multiple Integration and their Applications**		**15**	**CO5**
	5.1	Double integration: Introduction, Evaluation with given limits and over the given region		
	5.2	Change of order of integration, Change to polar form		
	5.3	Triple integration: Introduction, Evaluation with given limits		
	5.4	Triple integration using Cartesian, cylindrical and spherical coordinates		
	5.5	Application of Double integral & Triple integral : Area & Volume		
	#	**Self-learning topics**: Mass of lamina, Finding region of integration using any mathematical software		
		Total	**45**	

Students should prepare all self-learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in Tutorials.

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	B. S. Grewal	Higher Engineering Mathematics	Khanna Publications, India	43rd Edition 2014
2.	Shanti Narayan	A text book of Matrices	S. Chand, India	10th Edition 2004
3.	Erwin Kreyszig	Advanced Engineering Mathematics	Wiley Eastern Limited, India	10th Edition 2015
4.	Ramana B.V.	Higher Engineering Mathematics	Tata Mcgraw Hill New Delhi, India	34th Edition (reprint) 2019
5.	Glyn James	Advanced Modern Engineering Mathematic	Pearson Publication India	4th Edition 2010
6.	Gilbert Strang	Linear Algebra and Its Application	Brooks/Cole India, U.S.A	4th Edition 2005
7.	David C. Lay, Steven R. Lay, Judi J. McDonald	Linear Algebra and Its Application	Pearson, UK	6^{th} Edition 2022

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06E211</td><td colspan="6">Applied Science for Computer and Allied Programs (COMP/IT/AI&DS/CSBS/EXCP/CCE)</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>03</td><td colspan="2">--</td><td colspan="2">--</td><td>03</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td colspan="2">--</td><td colspan="2">--</td><td>03</td></tr>
<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td>100</td></tr>
</table>

Course prerequisites:

Basic knowledge of Physics at +2 or First Year Engineering level (mechanics, waves, electricity & magnetism). Understanding of basic electronic concepts such as current, voltage, resistance, and Ohm's law. Familiarity with mathematics (algebra, calculus, and differential equations) for solving physical models.

Course Objectives:

To introduce the fundamental principles of solid-state and semiconductor physics as applied to modern electronic devices. Explain the working of semiconductor and optoelectronic devices and their role in computing and communication technologies. Provide an understanding of the physics of sensors, MEMS/NEMS, and their applications in smart electronic systems. Familiarize students with emerging fields like spintronics, nanomaterials, and their impact on future electronics and computing. Develop conceptual clarity in quantum mechanics applied to nanoelectronics and introduce the basics of quantum computing. Enable students to relate physical concepts to real-world engineering applications in computer, electrical, and communication engineering.

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1: Demonstrate working principles of diodes, transistors and their application as solar cells, photovoltaics

CO2: Explain the concept of quantum confinement effect and quantum computing.

CO3: Understand the concept of electrostatics, magnetostatics and electromagnetic wave theory

CO4: Demonstrate the working principle of the energy storage devices and sensor materials.

CO5: Understand and explain the fundamentals of chip fabrication techniques.

Module No.	Unit No.	Details	Hrs.	CO
1	**Electronics & Optoelectronic Devices**		**09**	**CO1**
	1.1	**Didoes and Transistors**: p-n junction at equilibrium and under bias, diode equation, rectifier diode, Zener diode and Schottky diode characteristics, transistor biasing and static characteristics in CB, CE and CC modes, operating points, transistor as a switch, MOS structure, charge storage and flash memory device		
		LEDs, laser didoes, photodetectors, solar cells, Organic LEDs, (OLEDs), Quantum Dot LEDs (QELDs)		
2	**Applications of Quantum Mechanics**		**08**	**CO2**
	2.1	**Quantum Effects**: particle confinement and its effects on electronic and optical properties, potential barrier and quantum tunneling, nanomaterials for quantum hardware		
	2.2	**Quantum Computing Concepts:** Qubits and its physical realization**,** advantages over classical computers, quantum effects in computing - superposition, probability distribution and entanglement, illustrative quantum algorithm		
3	**Elements of Electromagnetic Wave Theory**		**07**	**CO3**
	3.1	**Electrostatics and Magnetostatics**: Electric field, Gauss's law, Electric potential and fields in conductors & dielectrics, Magnetic fields due to currents, Faraday's law of electromagnetic induction, Displacement current, Maxwell's equations		
	3.2	**Applications**: Electromagnetic wave propagation, polarization, Application of EMT in Wi-Fi, antennas, Bluetooth		
4	**Chemistry for Sensing and Storage Devices**		**11**	**CO4**
	4.1	**Energy Storage Devices and Applications:** Solid-state batteries: idea and advantages, Supercapacitors for quick charging, Heat management for data centres		
	4.2	**Sensor Materials and their applications:** Types of Sensors: resistive, capacitive, inductive, optical, piezoelectric, Sensor materials: piezo ceramics, strain-gauge materials, photodiode materials, MEMS silicon Basic sensor applications in embedded/IoT systems		
5	**Chemistry Behind Computers: From Chips to Storage**		**10**	**CO5**
	5.1	**Band Gap Engineering**: Band gap fundamentals and material categories, Basic material properties needed for device fabrication: mobility, resistivity, dielectric constant, fabrication of chips and microprocessors, Photoresists in lithography		
	5.2	**Materials for Data Storage**: Thin-film, and substrate materials used in ICs and MEMS, Magnetic storage (hard drives), Spintronic and giant magnetoresistance		
		Total	**45**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
7.	S. Salivahanan, N. Suresh Kumar	Electronic Devices and Circuits	McGraw Hill India	5th Edition, 2022
8.	Sheila Prasad, Hermann Schumacher, Anand Gopinath	High-Speed Electronics and Optoelectronics: Devices and Circuits	Cambridge University Press	1st Edition, 2009
9.	David J. Griffiths	Introduction to Quantum Mechanics	Pearson Education	2nd Edition, 2018
10.	Chris Bernhardt	Quantum Computing for Everyone	MIT Press	1st Edition, 2019
11.	Errol G. Lewars	Computational Chemistry: Introduction to the Theory and Applications of Molecular and Quantum Mechanics	Springer International Publishing	4^{th} Edition, 2024

<table>
<tr><th colspan="2">Course Code</th><th colspan="5">Course Title</th></tr>
<tr><td colspan="2">316U06E212</td><td colspan="5">Applied Science for Electronics and Allied Programs (VLSI/EXTC)</td></tr>
<tr><td colspan="2" rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td>PRACT</td><td>TUT</td><td colspan="2">Total</td></tr>
<tr><td>03</td><td>--</td><td>--</td><td colspan="2">03</td></tr>
<tr><td colspan="2">Credits Assigned</td><td>03</td><td>--</td><td>--</td><td colspan="2">03</td></tr>
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<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td>100</td></tr>
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Course prerequisites:

Basic understanding of Physics at +2 / First Year Engineering level (electricity, magnetism, and waves), Familiarity with semiconductor basics (conductors, insulators, band concept), Elementary mathematics (calculus, differential equations) for solving physics models.

Course Objectives:

Introduce the fundamental concepts of semiconductor physics and devices such as diodes, BJTs, MOSFETs, and solar cells. Provide understanding of quantum tunneling and photoelectric effect relevant to modern device physics. Develop knowledge of electromagnetic theory including Maxwell's equations and their role in wave propagation. Explain the properties of electrical and magnetic materials, superconductivity, and their technological applications. Familiarize students with the physics of sensors and transducers and their applications in electrical, electronic, and environmental systems. Relate physics principles to real-world applications in power systems, electronics, communication, and sensing technologies.

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1: Apply diode and transistor configurations in basic electronics circuits

CO2: Relate the Physics behind working of special-purpose and optoelectronic semiconductor devices

CO3: Apply fundamental laws of electrostatics and magnetostatics to electromagnetic waves

CO4: Explain the chemistry behind semiconductor materials, their properties and fabrication process

CO5: Understand chemistry of batteries, sensor materials and their applications

Module No.	Unit No.	Details	Hrs.	CO
1	**P-N Junctions**		**09**	**CO1**
	1.1	**P-N Junction diode**: Diode equation, application as rectifier: half wave, full wave, bridge rectifier and their parameters (ripple factor, rectification efficiency, transformer utilization factor), LC filter circuits, application of diode in clipper and clamper circuits		
	1.2	**Bipolar Transistor:** Common-Base (CB), Common Emitter (CE) and Common-Collector (CC) configurations and their characteristics, operating points, BJT-CE configuration as voltage amplifier and electronic switch, relay driver and LED driver circuits		
2	**Special-Purpose Diodes and Optoelectronics Devices**		**07**	**CO2**
	2.1	Zener diode, Schottky diode, tunnel diode		
	2.2	LEDs, laser diodes, photodiodes, solar cells, organic LEDs (OLEDs), quantum dot LEDs (QLEDs)		
3	**Elements of Electromagnetic Wave Theory**		**08**	**CO3**
	3.1	**Electrostatics**: Coulomb's law, Gauss's law, electric potential, field in conductors/dielectrics		
	3.2	**Magnetostatics**: Biot–Savart law, Ampere's law, Faraday's law of electromagnetic induction, Maxwell's equations, displacement current, one dimensional electromagnetic wave equation, speed of light		
	3.3	**Applications:** Wireless communication, Wi-Fi and Bluetooth signal transmission, wireless energy transfer based on Faraday's law, electromagnetic braking and levitation		
4	**Materials Chemistry for Semiconductors**		**11**	CO4
	4.1	**Bandgap Engineering**: Refractive index & optical absorption in semiconductor materials, defects in solids: vacancies, interstitials, dislocations**,** and their impact on device performance, doping materials and diffusion mechanisms		
	4.2	**Engineering Materials**: Thin-film, and substrate materials used in ICs and MEMS, Basic material properties needed for device fabrication: mobility, resistivity, dielectric constant, Deposition techniques, surface engineering		
	4.3	**Applications:** Magnetic storage (hard drives), Spintronic and giant magnetoresistance		
5	**Chemistry for Sensing and Storage Devices**		**10**	**CO5**
	5.1	**Energy Storage Devices and Applications:** Basic working principle of Lithium ion batteries, solid-state batteries: idea and advantages, Supercapacitors for quick charging, Heat management for data centres		
	5.2	**Sensors:** Basics of Sensors and their parameters, review of different types of sensors used in IOT**,** categories of sensors: resistive, capacitive, inductive, optical and piezoelectric		
	5.3	**Sensor material System**: Piezoelectric ceramics, strain-gauge materials, photodiode materials, mems-grade silicon applications of sensors in embedded and IOT platforms		
		Total	**45**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
12.	S. M Sze	Physics of Semiconductor Devices	John Wiley	7th Edition 2022
13.	Ben Streetman	Solid State electronic Devices	EEE	4th Edition, 2023
14.	S. Salivahanan, N. Suresh Kumar	Electronic Devices and Circuits	McGraw Hill India	5th Edition, 2022
15.	Sheila Prasad, Hermann Schumacher, Anand Gopinath	High-Speed Electronics and Optoelectronics: Devices and Circuits	Cambridge University Press	1st Edition, 2009
16.	Errol G. Lewars	Computational Chemistry: Introduction to the Theory and Applications of Molecular and Quantum Mechanics	Springer International Publishing	4th Edition, 2024
17.	Sushil Kumar Kashaw (Editor), Anshu man Dixit (Editor), Shivangi Agarwal	Bioinformatics, Computational Chemistry and AI in Drug Innovation	Routledge / CRC Press	1st Edition, 2025

Course Code	Course Title			
316U06E221	Engineering Mechanics			
Teaching Scheme (hrs.)	TH	PRACT	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Examination Scheme	Marks					
	CA		ESE (Theory/On-Screen)	PR/OR Exam	LAB/TUT CA	Total
	IA	MSE				
	20	30	50	--	--	100

Course prerequisites:

- Basics of units and conversions
- Basics of Trigonometry
- Newton's Laws of Motion

Course Objectives:

Engineering mechanics is the application of physics to solve problems involving common engineering elements. This course introduces system of forces and its effect on stationary and moving objects. The goal of this course is to expose students to problems in real-world scenarios and respond accordingly.

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1: Evaluate resultant and moment of a force system.
CO2: Analyze the equilibrium of force systems using free body diagram**.**
CO3: Study and analyze applications of equilibrium of force systems.
CO4: Explore the concept of kinematics of particle
CO5: Analyze the dynamic system using D'Alembert, work energy and impulse momentum principle.

Module No.	Unit No.	Details	Hrs.	CO
1		**System of forces**	**10**	**CO1**
	1.1	System of coplanar forces: Resultant of concurrent forces, parallel forces, non-concurrent non parallel system of forces, moment of force about a point, couples, Varignon's theorem, Principle of transmissibility of forces, Introduction to forces in space.		
	1.2	Centroid of wires/rods, Centroid of plane laminas: Plane lamina consisting of primitive geometrical shapes.		
2		**Equilibrium of Forces**	**08**	**CO2**
	2.1	Equilibrium of system of coplanar forces: Condition of equilibrium for concurrent forces, parallel forces and non-concurrent, non-parallel force system (general force system), Free body diagram.		
	2.2	Types of supports, loads, beams, determination of reactions at supports for various types of loads on simple beams.		
3		**Application of Equilibrium of Force Principle**	**10**	**CO3**
	3.1	Truss: Plane truss, zero force members, Analysis of plane truss by using method of joints and method of sections.		
	3.2	Friction: Laws of friction, cone of friction, angle of repose, equilibrium of bodies on inclined plane, application to problems involving wedges and ladders.		
4		**Kinematics of particle**	**08**	**CO4**
	4.1	Rectilinear motion: Variable motion, motion curves (a-t, v-t, s-t) (acceleration curves restricted to linear acceleration only).		
	4.2	Curvilinear motion: motion along plane curved path, velocity & acceleration in terms of rectangular components, tangential & normal component of acceleration.		
5		**Kinetics of particle**	**09**	**CO5**
	5.1	Newton's second law of motion, Force mass and acceleration, Introduction to basic concepts, equations of dynamic equilibrium, D' Alembert Principle, numerical problems.		
	5.2	Work energy principle: Work, Types of Energy i.e, kinetic energy, potential energy, strain energy, numerical problems.		
	5.3	Impulse and Momentum: Principle of linear impulse and momentum, law of conservation of momentum, impact and collision, direct central and oblique central impact.		
		Total	**45**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Tayal, A.K.	Engineering Mechanics, Statics and Dynamics	Universal Publication, India	14th Edition 2011
2	Bhavikatti S. S.	Engineering Mechanics	New Age international, India	Revised Edition 2019
3	Hibbeler, H. C. and Gupta	Engineering Mechanics, Statics and Dynamics	Prentice Hall Private limited, India	Revised Edition 2017
4	Bhattacharyya B.	Engineering Mechanics	Oxford University Press, India	2nd Edition 2014
5	Ram H.D. and Chauhan A.K.	Foundations and Applications of Engineering Mechanics	Cambridge University Press, UK	1st Edition 2015

Course Code	Course Title			
316U06C202	Digital Logic Design			
Teaching Scheme (hrs.)	TH	PTACT	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Examination Scheme	Marks					
	CA		ESE (Theory)	PR/OR Exam	LAB/TUT CA	Total
	IA	MSE				
	20	30	50	--	--	100

Course prerequisites:
None

Course Objectives:
Digital systems have a prominent role in everyday life. To understand the operation of digital systems, it is necessary to have a basic knowledge of digital circuits and their logical functions. The objective of this course is to familiarize the student with fundamental principles of digital design. It has been designed to address multidisciplinary students

Course Outcomes
At the end of the successful completion of the course, the student will be able to
- **CO1:** Understand the fundamentals of number systems, codes, and arithmetic operations.
- **CO2:** Represent, implement, and minimize logic functions using different techniques.
- **CO3:** Implement combinational logic circuits and their applications using basic gates and MSI devices.
- **CO4:** Implement sequential logic circuits and applications using MSI devices.
- **CO5:** Understand basic principles required for advanced Digital Design

Module No.	Unit No.	Details	Hrs.	CO
1	**Number Representation and Codes**		**09**	**CO1**
	1.1	Number Systems and Conversions: Binary, Octal, Decimal, Hexadecimal, Conversions between number systems, Signed numbers: Two's complement, Sign magnitude,		
	1.2	Binary Addition and Subtraction (1"s and 2"s complement method)		
	1.3	Binary-coded decimal (BCD), Gray code, Concept of parity for error detection.		
2	**Logic Representation and Minimization Techniques**		**10**	**CO2**
	2.1	Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR Universal Gates: Functionality of NAND and NOR. Rules and properties of Boolean algebra.		
	2.2	Combinational logic representation using truth table, sum of products (SOP) and products of sum(POS), Minimization techniques using Boolean Algebra, K-Map (2,3,4 Variable), Realization of circuits using Basic gates and Universal gates		
	#	**Self-learning topic**: Quine-McCluskey method for reduction of SOP/ POS expression		
3	**Combinational Circuits**		**12**	**CO3**
	3.1	Arithmetic Circuits: Study of combinational circuits such as Adder, subtractor, controlled adder/ subtractor, Use of MSI devices for adder		
	3.1	Multiplexer, demultiplexer, encoder, priority encoder, decoder, multiplexer tree and decoder tree., Use of MSI devices for multiplexer, decoder		
	3.2	1-bit and 2-bit magnitude comparator, 4 bit comparator		
	#	**Self-learning topic**: BCD adder/subtractor, Binary to 7 segment decoder		
4	**Sequential Logic Circuits**		**10**	**CO4**
	4.1	Latches and flip-flops: Types: SR, D, JK, T flip-flops truth tables, Excitation table, and characteristics equations.		
	4.2	Counters: Asynchronous counters, Synchronous counters, up/down counter, mod counters using flipflops, Use of MSI devices for counters		
	4.3	Registers: Shift registers using D-FF and types of shift registers (SISO, SIPO, PIPO, PISO). MSI shift register, Ring and Johnson Counter using Shift register.		
	#	**Self-learning topic**: Conversion of Flip-flop		
5	**Foundations of Digital Design**		4	CO5
	5.1	Introduction to Synchronous Machine (Mealy and Moore): Block diagram level. And its representation as state diagram.		
	5.2	Introduction, realization of logic expressions with PROM, PLA and PAL and their implementation.		
		Total	**45**	

Students should prepare all self-learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in laboratory.

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	R. P. Jain	*Modern Digital Electronics*	McGraw-Hill Education (India) Pvt Limited	4th Edition, 2008
2.	John M Yarbrough	*Digital Logic: Applications and Design*	Cengage Learning	2015
3.	A.P. Malvino and D.P.Leach	*Digital Principles and Applications*	Thomson Brooks/ Cole, India	10th Edition, 2015
4.	Morris Mano and Michael D. Ciletti	*Digital Design: With an Introduction to Verilog HDL*	Pearson Education, India	5th Edition, 2013
5.	A. Anand Kumar	*Fundamentals of Digital Circuits*	PHI Learning Pvt. Ltd	4th Edition, 2016

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06C203</td><td colspan="6">Environmental Science</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>02</td><td colspan="2">--</td><td colspan="2">--</td><td>02</td></tr>
<tr><td>Credits Assigned</td><td>02</td><td colspan="2">--</td><td colspan="2">--</td><td>02</td></tr>
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<table>
<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>50</td><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td></tr>
</table>

Module No.	Unit No.	Details	Hrs.	CO
1	**Humans and the Environment**		03	--
		• Human–environment interaction: hunter-gatherers, fire, agriculture, city-states, ancient civilizations & Indic knowledge of sustainability • Middle Ages, Renaissance, Industrial Revolution, population growth & global environmental change • Environmental ethics & environmentalism: Anthropocentric vs Ecocentric, Club of Rome, Stockholm 1972, WCED & Rio Summit		
2	**Natural Resources & Sustainable Development**		04	--
		• Concept & classification of natural resources (biotic/abiotic, renewable/non-renewable) • Biotic resources: forests, grasslands, wetlands, wildlife, aquatic resources; microbes as resources • Water, soil & mineral resources: exploitation, challenges, conflicts • Energy resources (conventional & non-conventional) + Introduction to Sustainable Development & SDGs		
3	**Environmental Issues: Local, Regional & Global**		04	--
		• Environmental issues & scales: micro–planetary, local/regional/global phenomena • Types of pollution: air, water, noise, soil, thermal, radioactive; hazardous & municipal solid waste • Land use/cover change: deforestation, desertification, urbanization • Global change: ozone depletion, climate change, natural & anthropogenic disasters		
4	**Conservation of Biodiversity & Ecosystems**		04	--
		• Biodiversity: levels, types, hotspots, threat categories • Ecosystems & ecosystem services: forests, wetlands, grasslands, agriculture, coastal & marine • Threats to biodiversity: land-use change, commercial exploitation, invasives, disasters, climate change • Conservation policies: in-situ, ex-situ, protected areas, traditional knowledge, gender & conservation		

5	**Environmental Pollution & Health**		**04**	**--**
	5.1	• Pollution concepts: sources, assimilative capacity, point & non-point • Air pollution: sources, primary/secondary pollutants, criteria pollutants, indoor air pollution, health impacts • Water & soil pollution: sources, quality parameters, health impacts; solid & hazardous waste • Noise, thermal & radioactive pollution: definitions, sources, standards, human/ecosystem impacts		
6	**Climate Change – Impacts, Adaptation & Mitigation**		04	**--**
		• Understanding climate change: natural variability, GHG emissions, projections, 1.5°C & 2°C targets • Climate change impacts: sea level rise, ecosystems, agriculture, health, urban infrastructure • Vulnerability & adaptation: resilience, climate-resilient development, indigenous knowledge • Mitigation strategies: GHG reduction, carbon neutrality, policies, renewable energy, INDCs & climate justice		
7	**Environmental Management**		04	**--**
		• Environmental laws: Constitutional provisions (Art. 48A, 51A(g)), Forest/Wildlife/Pollution control acts • Environmental management systems: ISO 14001, EMS concepts • Circular economy, life cycle analysis, cost-benefit analysis • Environmental audit, risk assessment, 3Rs, eco-labelling & sustainable waste management		
8	**Environmental Treaties & Legislation**		03	**--**
		• International cooperation: agreements, conventions, protocols, COP mechanism • Major treaties: CBD, CITES, Ramsar, Montreal, Kyoto, Paris, Basel, Stockholm, UNFCCC • Indian legislations & institutions: Wildlife Act, EPA 1986, Biodiversity Act 2002, NGT, MoEFCC, UNEP, IPCC		
9	**Field Work (60) hours / Field Immersion Activities (Nature Basket)**		--	**--**
		• additional 2 credits to be covered		
		Total	**30**	

Course Code	Course Title					
316U06C204	Object-Oriented Programming Methodology					
Teaching Scheme (hrs.)	TH	PRACT	TUT	Total		
	02	02	--	04		
Credits Assigned	02	01	--	03		
Examination Scheme	Marks					
	CA		ESE (On-Screen*)	PR/OR Exam	LAB/TUT CA	Total
	IA	MSE				
	--	--	50	--	50	100

*The End Semester Examination (ESE) will be conducted in an onscreen format and may include a combination of theoretical questions, practical coding tasks, and/or an oral/viva component

Course prerequisites:

Students should have basic knowledge of:

- Programming fundamentals
- Data types, loops, control structures
- Basic understanding of algorithms and flowcharts

Course Objectives:

- Understand the principles and paradigms of object-oriented programming.
- Learn the structure and relationships of classes, objects, and methods.
- Apply concepts of inheritance, polymorphism, abstraction, and encapsulation in software development.
- Explore exception handling, multithreading, and file management in an object-oriented environment.

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1. Understand the fundamentals of object-oriented programming and complex system design.
CO2. Identify and apply classes, objects, methods, and constructors to develop modular programs.
CO3. Implement advanced OOP concepts such as inheritance, polymorphism, and abstract classes.
CO4. Implement applications involving multithreading, exception handling, file operations, and reusable components.

Module No.	Unit No.	Contents	Hrs.	CO
1	**Fundamentals of Object-Oriented Programming**		05	CO1
	1.1	Introduction, Basic Program Construction, Procedural Programming Approach, Structured Programming Approach, Modular Programming Approach, OOP Approach		
	1.2	The Structure of Complex Systems with attributes, The Inherent Complexity of Software		
2	**Classes and Objects, methods and constructors**		08	CO2
	2.1	The Object Model, Elements of the Object Model – (Features) Applying the Object Model		
	2.2	The Nature of an Object, Relationships among Objects, The Nature of a Class Relationships among Classes The Interplay of Classes and Objects		
	2.3	Identifying Classes and Objects Key Abstractions and Mechanisms		
	2.4	Member, method, State of an object. Memory allocation of an object		
3	**Inheritance and Polymorphism**		07	CO3
	3.1	Inheritance, Types of Inheritance		
	3.2	Method overloading & overriding, constructor, Types of constructors, Constructor Overloading, Concept of memory management of unused objects/references (destructor /Garbage collection), Early vs. Late Binding, Runtime polymorphism		
	#	**Self-Learning topic**: Template /Generic classes		
4	**Packages/namespaces, Exception Handling, Multithreading, files**		10	CO4
	4.1	Introduction to Packages/Namespaces.		
	4.2	Introduction to Exception Handling.		
	4.3	Multithreading: Thread life cycle, Multithreading advantages and issues, Simple thread program, Thread synchronization.		
	#	**Self-Learning topic**: The Standard Template Library, Various packages/ namespaces, File handling		
		Total	30	--

Note: Each theory topic should be supported by relevant programming examples delivering.

Recommended Books:

Sr. No	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Grady Booch, James Rumbaugh, Ivar Jacobson	*Unified Modeling Language*	Pearson Education	3rd Edition, 2007
2	Herbert Schildt	*Java The Complete Reference*	Tata McGraw-Hill	12th Edition, 2022
3	E Balagurusamy	*Object Oriented Programming in C++*	Tata McGraw Hill	5th Edition, 2011
4	Sheetal Taneja and Naveen Kumar	Python Programing: A Modular Approach	Pearson India	2nd Edition, 2018, India
5	Sachin Malhotra, Saurabh Chaudhary	Programming in Java	Oxford University, India	2nd Edition, 2018
6	NPTEL: C++	https://onlinecourses.nptel.ac.in/noc21_cs02/preview		
7	NPTEL: Java	https://onlinecourses.nptel.ac.in/noc22_cs47/preview		
8	NPTEL: Python	https://nptel.ac.in/courses/106106145		

Note:
1. The total marks assigned for Continuous Assessment (LAB/TUT) as per scheme can be distributed in a number of components as given above
2. ***All components should be same across all divisions.***
3. The course coordinator should decide the rubrics/distribution of marks for different components in consultation with all other faculty teaching the same course.
4. ***The course department should decide the language of implementation as per the requirement of the programme.***
5. The rubrics/distribution should be uniform for all batches.
6. The rubrics/distribution should be communicated to all students at the beginning of the semester.
7. C++/Java/Python is recommended to teach
8. No IA, ISE, ESE on self-learning topics, but self-learning topics can be covered in laboratory experiments.

Course Code	Course Title			
316U06T201	**Presentation and Communication Skills**			
Teaching Scheme (hrs.)	**TH**	**PRACT**	**TUT**	**Total**
	01	--	01	02
Credits Assigned	01	--	01	02

Examination Scheme	**Marks**					
	CA		**ESE (Theory/On-Screen)**	**PR/OR Exam**	**LAB/TUT CA**	**Total**
	IA	**MSE**				
					50	50

Course prerequisites:
Following topics of higher secondary level English are required as prerequisites of this course:

- Grammatical proficiency
- Fundamental Linguistic Skills (LSRW)
- Letter Writing

Course Objectives: The focus of this course is to improve presentation and soft skills. The course aims to inculcate in students, self-management and interpersonal skills for enhanced workplace communication. The course also focuses on developing soft skills and business writing skills of the students.

Course Outcomes
At the end of the successful completion of the course, the student will be able to
CO1: Use basic communication and behavioral skills in day-to-day communication.
CO2: Present themselves effectively in business meetings and group discussions.
CO3: Perform confidently and effectively in campus placements.
CO4: Compose effective business letters.

Module No.	Unit No.	Details	Hrs.	CO
1		Soft Skills	**4**	**CO1**
	1.1	Non-verbal Communication		
	1.2	Barriers to communication		
	1.3	Emotional Intelligence		
2		Effective Business Presentations	**4**	**CO2**
	2.1	Business Meetings: Notice, Agenda, Minutes		
	2.2	Presentations: Language and Style		
	2.3	Group Discussion		
3		Employment Skills	**4**	**CO3**
	3.1	Interviews		
	3.2	Resume		
	3.3	Job Application		
4		Business Correspondence	**3**	**CO4**
	4.1	Quotations		
	4.2	Order Letters		
	4.3	Claims and Adjustments		
		Total	**15**	

Note: There will be 8-10 graded tutorials including group discussion activity throughout the semester

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
6.	Sharma, R C. and Krishna Mohan	*Basic Correspondence and Report Writing: A Practical Approach to Business and Technical Communication*	Tata McGraw-Hill Publishing Company Limited, India	2017
7.	Wallace, Harold R. & Ann Masters	*Personal Development for Life and Work.*	Cengage, USA	10^{th} Edition, 2012
8.	Sullivan, Jay	*Simply Said: Communicating Better at Work and Beyond*	Wiley publications	1^{st} Edition, 2018 (reprint)
9.	Petes S. J., Francis.	Soft Skills and Professional Communication.	Tata McGraw-Hill Education, India	First Edition 2011

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06L201</td><td colspan="6">Applied Science Laboratory - II</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>--</td><td colspan="2">02</td><td colspan="2">--</td><td>02</td></tr>
<tr><td>Credits Assigned</td><td>--</td><td colspan="2">01</td><td colspan="2">--</td><td>01</td></tr>
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<table>
<tr><td rowspan="4">Examination Scheme</td><td colspan="7">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
</table>

Note: List of experiments/tutorials, scheme for continuous assessment and rubrics will be shared by course coordinators from time to time at the beginning of every semester

- Applied Science laboratory will comprise of 5-6 experiments each from physics and chemistry concepts mapping to some of the COs. Combined grading will be applicable.
- A few experiments may be specific to the core branch/program

<table>
<tr><th>Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td>316U06L221</td><td colspan="6">Engineering Mechanics</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td colspan="2">PRACT</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>--</td><td colspan="2">02</td><td colspan="2">--</td><td>02</td></tr>
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<tr><td rowspan="4">Examination Scheme</td><td colspan="6">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
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Distribution of LABCA Marks*	
Instructions:	● LABCA includes journal / written record of experiment (25 marks) and oral based on experiments performed (25 marks). ● Experiments will be evaluated out of 25 marks each. There will be a minimum of 10 experiments to be performed in a semester. The <u>average</u> will be taken to determine marks out of 25. Oral will be taken at the end of semester for 25 marks. Total marks for LABCA will be 50. ● A student must complete the stipulated minimum number of experiments satisfactorily in order to clear the LABCA.
Rubrics (Journal):	1. Proper conclusion of experiment 2. Appropriate and neat sketches 3. Proper calculations 4. Timely submission 5. Attendance during conduction of practical

<table>
<tr><th>Course Code</th><th colspan="5">Course Title</th></tr>
<tr><td>316U06L202</td><td colspan="5">Digital Logic Design Laboratory</td></tr>
<tr><td rowspan="2">Teaching Scheme (hrs.)</td><td>TH</td><td>PTACT</td><td>TUT</td><td colspan="2">Total</td></tr>
<tr><td>--</td><td>02</td><td>--</td><td colspan="2">02</td></tr>
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<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>--</td><td>--</td><td>50</td><td>50</td></tr>
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Lab CA will comprise of a variety of components such as quizzes, onscreen exam, viva voce, journal, GDs etc. covering entire syllabus of "Digital Logic Design" (316U06C202). Details will be shared by course teachers at the beginning of every semester.

<table>
<tr><th colspan="2">Course Code</th><th colspan="6">Course Title</th></tr>
<tr><td colspan="2">316U06L203</td><td colspan="6">Maker Space Laboratory - II</td></tr>
<tr><td colspan="2" rowspan="2">Teaching Scheme (hrs.)</td><td colspan="2">TH</td><td colspan="2">PRACT</td><td>TUT</td><td>Total</td></tr>
<tr><td colspan="2">02</td><td colspan="2">02</td><td>--</td><td>02</td></tr>
<tr><td colspan="2">Credits Assigned</td><td colspan="2">--</td><td colspan="2">01</td><td>--</td><td>01</td></tr>
<tr><td rowspan="4">Examination Scheme</td><td colspan="7">Marks</td></tr>
<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">Project*</td><td colspan="2" rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>--</td><td>--</td><td>--</td><td>50</td><td colspan="2">50</td><td>50</td></tr>
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*Project related to Project Based Learning Lab

Course prerequisites:
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Course Objectives:
The main objective of the **Maker Space Laboratory - II** is to provide all engineering students with theoretical and practical knowledge of the manufacturing environment. This course is the foundation of the real industrial environment, and it helps students develop and improve relevant technical hand skills. It teaches the fundamentals of various hand tools, power tools, machine tools, and their applications in various areas of manufacturing.

Course Outcomes

CO1: Understand the functions and applications of various tools, interpret job drawings, plan processes, and perform operations to produce basic components from raw materials.
CO2: Apply the safety measures practiced while using the tools, equipment, devices, etc.
CO3: Understand the entire product development and manufacturing process, collaborate as a team, learn new skills, and develop leadership abilities.\
CO4: Build products on their own, maintain dimensional accuracy, and boost creativity by exploring new ideas.
CO5: Apply design thinking principles to solve real world problems in a team

Note:
1. Each student must complete three shops during semester II as specified in the table
2. There will be 4-5 students per team. Max 10 hrs./team to complete assigned task in each shop
3. Assessment:
   i. Continuous assessment will be done throughout the semester for the allotted shops
   ii. Quality of finished product
4. For PBL module, there will be group project. Students need to work in a group of 4-5 based on the activities/training done in semester I. Project should preferably address Environment and Sustainability related problem solution (refer to 17 SDGs)

Shop	Department/Program
Drone Technology Shop	Compulsory for All Programs
Project Based Learning (PBL)	Compulsory for All Programs
Printed Circuit Board (PCB) Shop	Computer Engineering, Information Technology
Computer Hardware and Assembly Shop	Electronics Engineering, Electronics and Telecommunication Engineering
Fabrication Process Shop*	Mechanical Engineering
Woodcraft and Fabrication Shop*	Mechanical Engineering

*One of the two workshops based on the shop allotted in semester I

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